

Heart Failure: Observation Care

OCG: OC-024 (ISC)

[Link to Codes](#)

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Observation Care Admission Criteria

Note: For patients requiring inpatient admission rather than Observation care, see Heart Failure [ISC](#).

Note: For patients with clinically active secondary conditions, see Heart Failure Multiple Condition Management Guidelines.

- Observation care is indicated for **1 or more** of the following(1)(2)(3)(4)(5):
 - Vital sign abnormality
 - Pulmonary edema[A]
 - Hypoxemia
 - Tachypnea
 - Dyspnea
 - Altered mental status
 - Abnormal electrolytes (eg, hypokalemia, hyponatremia)
 - Change in renal function (reduction in estimated glomerular filtration rate from baseline)  eGFR - Adult Calculator

Observation Care Discharge Criteria

- Discharge to non-acute-care follow-up is appropriate for patient with **ALL** of the following(1)(2)(3)(4)(5):
 - Hemodynamic stability
 - Tachypnea absent
 - No evidence of acute myocardial ischemia
 - Dyspnea absent, at baseline, or acceptable for next level of care
 - Oxygen saturation greater than 90% on room air, or at baseline supplemental oxygen
 - Pulmonary edema absent, at baseline, or acceptable for next level of care
 - Mental status at baseline
 - Cardiac rhythm acceptable for next level of care (eg, normal sinus rhythm, atrial fibrillation with controlled heart rate)
 - Peripheral or sacral edema absent, at baseline, or acceptable for next level of care
 - Renal function at baseline, or acceptable for next level of care
 - Electrolytes acceptable for next level of care (eg, serum sodium and potassium)
 - Volume status at baseline, or acceptable for next level of care
 - Any precipitating cause for acute decompensation identified and addressed (eg, dietary or medication nonadherence)
 - Left ventricular function is known, or evaluation is planned after discharge.
 - Education regarding activity, diet, medications, follow-up appointments, home weight monitoring, and instructions for worsening symptoms have been provided.
 - Medication regimen reviewed and optimized (eg, diuretics, renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor (ARNI) preferred over ACE inhibitor or ARB), aldosterone antagonist, beta blockade, sodium-glucose cotransporter-2 inhibitor (eg, dapagliflozin, empagliflozin, sotagliflozin), vericiguat, or vasodilator (eg, hydralazine, isosorbide)) [A][B](1)(7)(8)(9)
 - Social Determinants of Health Assessment completed (as indicated)
 - Adequate evaluation completed (as indicated) and negative for condition requiring inpatient admission. Evaluation may include:

- Cardiac biomarkers, including Troponin, B-type natriuretic peptide (BNP), or N-terminal pro-B-type natriuretic peptide (NT-proBNP) level(10)(11)
- CBC, electrolytes, BUN, creatinine, digoxin level, liver function tests, other laboratory tests
- ECG
- Chest x-ray
- Pulmonary ultrasound(12)
- ABGs
- Pulse oximetry
- Cardiac monitoring
- Echocardiogram
- Radionuclide ventriculography
- Pulmonary CT angiogram or lung ventilation perfusion scan
- Cardiac MRI
- Exercise testing (eg, 6-minute walk or maximum cardiopulmonary exercise test)
- Stress echocardiogram or nuclear stress-test imaging(13)
- Pulmonary function tests
- Cardiology consultation

Evidence Summary

Criteria

The evidence for the clinical indications found in this guideline includes 2 published peer reviewed articles, 2 specialty society or other evidence-based guidelines, and 1 book section.

Rationale

Use of this MCG care guideline helps the clinician identify patient-specific clinical factors that make it reasonable to admit a patient to observation care to continue to receive medically necessary hospital care rather than be prematurely discharged from the emergency department. This continued hospital care is needed before suitability for inpatient care can be determined. The duration of observation care is finite and short. For example, for Medicare enrollees, the determination to admit a patient to inpatient care or discharge the patient to a lower level of care is made before the patient spends a second midnight receiving medically necessary hospital care (ie, satisfies CMS' Two-Midnight Rule).

Use of these evidence-based clinical criteria to support decision making around the need for observation treatment is of clinical benefit to the patient and is crucial for quality patient care. Use of evidence-based criteria ensures patients receive the most appropriate care for their specific condition, reduces unnecessary risks, promotes recovery, and provides a standard that can reduce disparities in care. Not only do evidence-based criteria help align treatment decisions with best practice, but they can also help reduce unwarranted variation in care by fostering consistency and equality in healthcare provision across regions and facilities.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 412.3(14); 42 CFR 419.22(15); 42 CFR 422.101(16)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 6 - Hospital Services Covered Under Part B(17); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(18); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(19)

Medicare Coverage Determinations: Medicare Coverage Database(20)

References

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Footnotes

[A] A specialty clinical guideline, based upon a systematic review of the evidence, concludes that heart failure patients with evidence of significant fluid overload should be promptly treated with intravenous loop diuretics (eg, initial dose twice the home regimen in terms of oral to parenteral milligram equivalents) and titrated to resolution of clinical signs and symptoms of congestion (eg, serial doubling of dosage as bolus or continuous infusion).(1) This guideline further recommends increasing the diuretic dose or adding a second diuretic (eg, thiazide diuretic or aldosterone antagonist with diuretic properties) until an effective regimen is identified that is adequate to relieve congestion.(1) [A in Context Link 1, 2]

[B] A specialty guideline recommends 4 medication classes as guideline-directed medical therapy (GDMT) for patients with heart failure with reduced ejection fraction (HFrEF): beta-blockers, renin-angiotensin inhibitors (angiotensin receptor-neprilysin inhibitor preferred), aldosterone blockers, and sodium-glucose cotransporter-2 inhibitors.(1) This same guideline recommends the following for patients hospitalized with HFrEF absent clear contraindications: pre-existing GDMT should be continued and optimized (titration to therapeutic dose begun as inpatient), GDMT should not be discontinued due to mild decrease in renal function or blood pressure, GDMT should be initiated while hospitalized when clinical stability achieved (if all drug classes not on board), and if GDMT is discontinued it should be reinstated as soon as possible.(1) Vericiguat further reduces the risk of adverse cardiac outcomes in heart failure and may either be started with the standard quadruple therapy outlined above for severe worsening heart failure, used as a substitute agent for patients with contraindication or intolerance to one or more standard medications, or used as an add-on therapy for inadequate response to

optimal quadruple therapy.(5) An expert specialty society panel also recommends these 4 medication classes as GDMT for heart failure with preserved ejection fraction (HFpEF).(6) [B in Context Link 1]

Definitions

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
 - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
 - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
 - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
 - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
 - Coma (ie, not arousable)

References

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Clinically active asthma

- Clinically active asthma refers to asthma necessitating treatment while hospitalized with oral or parenteral (IV) corticosteroids.

Clinically active atrial fibrillation

- Clinically active atrial fibrillation refers to atrial fibrillation necessitating electrical cardioversion or treatment while hospitalized with parenteral (IV) medication to control heart rate (eg, digoxin, diltiazem, metoprolol) or to cardiovert (eg, amiodarone, procainamide).

Clinically active chronic obstructive pulmonary disease

- Clinically active chronic obstructive pulmonary disease (COPD) refers to COPD necessitating assisted ventilation (eg, BPAP, CPAP) or treatment while hospitalized with oral or parenteral (IV) corticosteroids.

Clinically active diabetes

- Clinically active diabetes refers to diabetes necessitating treatment with insulin while hospitalized.

Clinically significant dementia

- Clinically significant dementia refers to dementia necessitating treatment while hospitalized with antipsychotic medication (eg, haloperidol, risperidone) or dementia-specific medication (eg, donepezil, memantine).

Clinically significant malnutrition

- Clinically significant malnutrition refers to patients presenting with findings indicative of poor nutritional status; examples include unintentional weight loss, BMI less than 18.5, inadequate nutritional intake, loss of subcutaneous fat (eg, orbital, triceps, over ribs), or loss of muscle mass (eg, temples, clavicles, hands).  BMI Calculator

Heart Failure Multiple Condition Management Guidelines

- For patients who also have:
 - Clinically active asthma, see Heart Failure with Clinically Active Asthma MCM
 - Clinically active atrial fibrillation, see Heart Failure with Clinically Active Atrial Fibrillation MCM
 - Clinically active chronic obstructive pulmonary disease and Clinically active diabetes, see Heart Failure with Clinically Active Chronic Obstructive Pulmonary Disease and Clinically Active Diabetes MCM
 - Clinically active chronic obstructive pulmonary disease, see Heart Failure with Clinically Active Chronic Obstructive Pulmonary Disease MCM
 - Clinically active diabetes, see Heart Failure with Clinically Active Diabetes MCM
 - Clinically significant dementia, see Heart Failure with Clinically Significant Dementia MCM
 - Clinically significant malnutrition, see Heart Failure with Clinically Significant Malnutrition MCM
 - Severe renal failure, see Heart Failure with Severe Renal Failure MCM

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
 - Hemodynamic abnormalities at baseline or acceptable for next level of care

- Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
 - Tachycardia absent
 - Hypotension absent
 - No evidence of inadequate perfusion (eg, no myocardial ischemia)
 - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

References

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5. Singer M, et al. The Third International Consensus definitions for sepsis and septic shock (Sepsis-3). Journal of the American Medical Association 2016;315(8):801-810. DOI: 10.1001/jama.2016.0287.

Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4):
 - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
 - Low blood pressure, as indicated by **1 or more** of the following:
 - SBP less than 90 mm Hg in adult or child 10 years or older[A]
 - Mean arterial pressure[A][B] less than 70 mm Hg in adult or child 10 years or older
 - Decrease in baseline mean arterial pressure of 25% or more,[A][B] with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)
 - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age[A]
 - SBP less than 70 mm Hg in infant 1 to 11 months of age[A]

References

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Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP. A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hypotension absent

- Hypotension absent[A], as indicated by **1 or more** of the following(1)(2)(3)(4):
 - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
 - Mean arterial pressure[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
 - Mean arterial pressure[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
 - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
 - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

References

1. Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
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Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = $1/3 \text{ SBP} + 2/3 \text{ DBP}$.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following(1):
 - Patient without baseline need for supplemental oxygen with oxygen saturation (SaO₂) of less than 90% or arterial blood gas partial pressure of oxygen (PO₂) of less than 60 mm Hg (8.0 kPa) on room air[A][B]
 - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SaO₂ greater than 89% or PO₂ greater than 59 mm Hg (7.9 kPa)[A]
 - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

References

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Footnotes

- A. Pulse oximetry (SpO₂) may underestimate arterial saturation (SaO₂), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.(2) Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.(2) Specific target values may require adjustment when care is delivered at high altitude.(1)(3)
- B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Observation care

- When a Clinical Indication says "...despite (or beyond) observation care," the intent is to signify that, despite an attempt to address a sign, symptom, finding, or treatment need in observation care, a continued need for hospital-based care persists beyond the observation care time frame (ie, Observation Care Discharge Criteria are not met). The time frame of observation care is determined by regulation (eg, CMS's Two-Midnight Rule) or payer-provider contractual agreement. MCG's understanding of observation care is that it must be finite and of short duration. The Two-Midnight Rule for Medicare patients (inpatient is defined as medical necessity for hospital-based care across 2 or more midnights) provides a good example of what is meant by a short duration (ie, observation care should not continue beyond the second midnight).

Orthostatic hypotension

- Orthostatic hypotension,^{[A][B]} as indicated by **1 or more** of the following⁽¹⁾⁽²⁾⁽³⁾:
 - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
 - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

References

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Footnotes

- A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.⁽¹⁾⁽²⁾⁽³⁾
- B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Severe renal failure

- Severe renal failure refers to renal failure or insufficiency necessitating inpatient dialysis or treatment with phosphate-binding agent medication (eg, sevelamer), or medication to treat secondary hyperparathyroidism (eg, cinacalcet).

Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health⁽¹⁾⁽²⁾⁽³⁾⁽⁴⁾:
 - Housing insecurity, as indicated by **1 or more** of the following:
 - Individual or caregiver's current living situation is **1 or more** of the following⁽⁵⁾:
 - Does not have own housing (eg, staying in a hotel, shelter, or with others)
 - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
 - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
 - Current housing has **1 or more** of the following:
 - Electrical appliances (eg, stove, refrigerator) not working or unavailable
 - Insufficient heating or cooling
 - Insufficient ventilation
 - Lead paint or pipes
 - Mold
 - Pests (eg, bugs) or rodents
 - Smoke detectors not working or unavailable
 - Food insecurity, as indicated by **1 or more** of the following⁽⁶⁾:
 - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
 - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
 - Insufficient transportation, as indicated by **1 or more** of the following⁽⁷⁾:
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
 - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
 - Insufficient utilities, as indicated by **1 or more** of the following⁽⁸⁾:
 - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
 - In the past year, electric, water, gas, or oil company threatened to shut off services.
 - Personal safety risk, as indicated by **2 or more** of the following⁽⁶⁾:
 - Individual is sometimes or frequently physically hurt by another person (including family member).
 - Individual is sometimes or frequently insulted or talked down to by another person (including family member).

- Individual is sometimes or frequently threatened with physical harm by another person (including family member).
- Individual is sometimes or frequently screamed or cursed at by another person (including family member).
- Insufficient dependent care, as indicated by **1 or more** of the following:
 - In the past year, individual or caregiver was unable to work due to lack of dependent care.
 - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
 - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
- Depression risk, as indicated by **ALL** of the following:
 - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
 - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

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Tachycardia

- Tachycardia,[A][B] as indicated by **1 or more** of the following(1)(2):
 - Heart rate greater than 100 beats per minute in adult or child age 6 years or older
 - Heart rate greater than 115 beats per minute in child 3 to 5 years of age
 - Heart rate greater than 125 beats per minute in child 1 or 2 years of age
 - Heart rate greater than 130 beats per minute in infant 6 to 11 months of age
 - Heart rate greater than 150 beats per minute in infant 3 to 5 months of age
 - Heart rate greater than 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. *Principles and Practice of Hospital Medicine*. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. *The Harriet Lane Handbook: A Manual for Pediatric House Officers*. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically

important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachycardia absent

- Tachycardia^{[A][B]} absent, as indicated by **1 or more** of the following(1)(2):
 - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
 - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
 - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
 - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
 - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
 - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
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Tachypnea

- Tachypnea^{[A][B]} as indicated by respiratory rate of **1 or more** of the following(2)(3):
 - Greater than 18 breaths per minute in adult or child 13 years of age or older^[A]
 - Greater than 22 breaths per minute in child 6 to 12 years of age^[A]
 - Greater than 25 breaths per minute in child 3 to 5 years of age^[A]
 - Greater than 30 breaths per minute in child 1 or 2 years of age^[A]
 - Greater than 40 breaths per minute in infant 6 to 11 months of age^[A]
 - Greater than 45 breaths per minute in infant 3 to 5 months of age^[A]
 - Greater than 60 breaths per minute in infant 1 or 2 months of age^[A]

References

1. Schrager DL. Approach to the patient with abnormal vital signs. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:28-31.e2.
2. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
3. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

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deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

- B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath to breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.⁽¹⁾ This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.⁽¹⁾

Tachypnea absent

- Tachypnea^{[A][B]} absent, as indicated by respiratory rate of **1 or more** of the following⁽²⁾⁽³⁾:
 - Less than or equal to 18 breaths per minute in adult or child 13 years of age or older
 - Less than or equal to 22 breaths per minute in child 6 to 12 years of age
 - Less than or equal to 25 breaths per minute in child 3 to 5 years of age
 - Less than or equal to 30 breaths per minute in child 1 or 2 years of age
 - Less than or equal to 40 breaths per minute in infant 6 to 11 months of age
 - Less than or equal to 45 breaths per minute in infant 3 to 5 months of age
 - Less than or equal to 60 breaths per minute in infant 1 or 2 months of age

References

1. Schrager DL. Approach to the patient with abnormal vital signs. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:28-31.e2.
2. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
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Troponin

- Troponin elevation alone does not indicate acute myocardial infarction; factors to consider include⁽¹⁾⁽²⁾:
 - Patients can have elevated levels of cardiac troponin due to other causes of myocardial injury, either acutely or chronically (eg, heart failure, myocarditis, pulmonary embolism, cardiomyopathy, defibrillator shock).
 - Patients with an elevated troponin should, on subsequent measurement, have a fall or rise in the troponin level of more than 20% to confirm acuity of myocardial injury. Absent this variation, the acuity of the myocardial injury should be questioned.
 - Attributing acute troponin release to myocardial ischemia depends on the presence of associated symptoms (eg, chest pain, dyspnea), ECG changes (eg, acute ST-segment deviation), and imaging findings (eg, new wall motion abnormalities on echocardiogram).
 - A report from a facility (eg, hospital) laboratory of an elevated troponin level, in general, indicates a level above the 99th percentile for the specific troponin assay used. The precise value that represents the 99th percentile depends on the specific troponin assay used.

- The recommended time interval between serial troponin level assessments (eg, to detect fall or rise in troponin level) is 1 to 3 hours for high-sensitivity troponin, and 3 to 6 hours for conventional (contemporary) troponin assays.
- Evidence-based specialty society guidelines recommend use of high-sensitivity troponin assays, rather than older less-sensitive assays (collectively labeled contemporary assays).(1)(3) This is due to more rapid detection or exclusion of myocardial injury and increased diagnostic accuracy with high-sensitivity assays.(1)(3)

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Vital sign abnormality

- Vital sign abnormality, as indicated by **ALL** of the following:
 - Vital sign findings not as expected for chronic patient condition or baseline (eg, intentionally low blood pressure in heart failure)
 - Vital sign abnormality, as indicated by **1 or more** of the following:
 - Tachycardia
 - Hypotension
 - Orthostatic hypotension

Codes

ICD-10 Diagnosis: I09.81, I11.0, I13.0, I13.2, I50.1, I50.20, I50.21, I50.22, I50.23, I50.30, I50.31, I50.32, I50.33, I50.40, I50.41, I50.42, I50.43, I50.814, I50.82, I50.83, I50.84, I50.89, I50.9, T86.22 [Hide]

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